

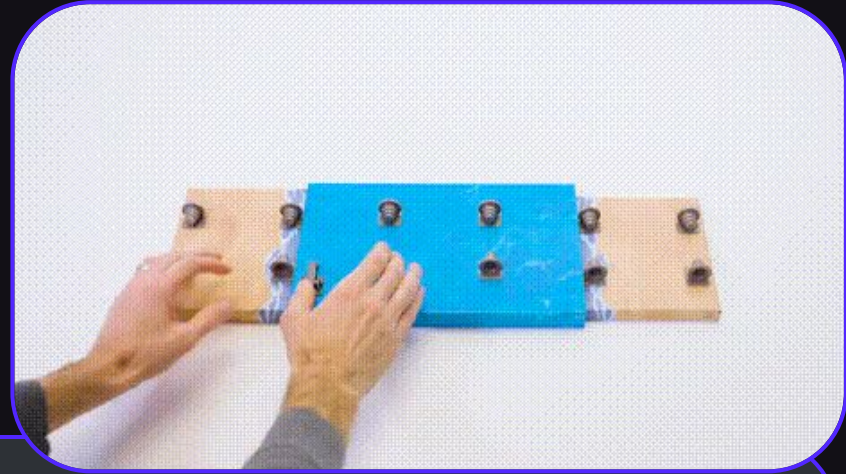
Bridge Mat

Updated Bridge or Bust's game mat to:

- Reduce friction when paper wrap is added
- Increase stability with large bridges
- Simplify the design without losing function
- Make manufacturability easier for the producer

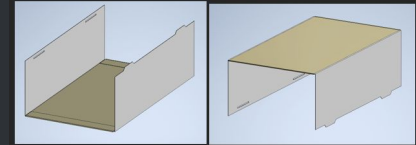


"I have not failed. I've just found 10,000 ways that won't work." - Thomas A. Edison

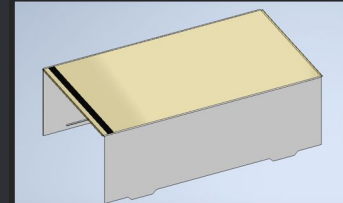


Learned how factory communication works and created a step by step guide to replicate my process. The guide is made to be efficient and repeatable for the factory.

Step 1: Fold the top and bottom flaps of BR_Mat_Center_Sleeve_Inside_Pattern_v2 inward, slide the plastic sheet onto BR_Mat_Center_Sleeve_Inside_Pattern_v2. Fold the left and right edges inward.

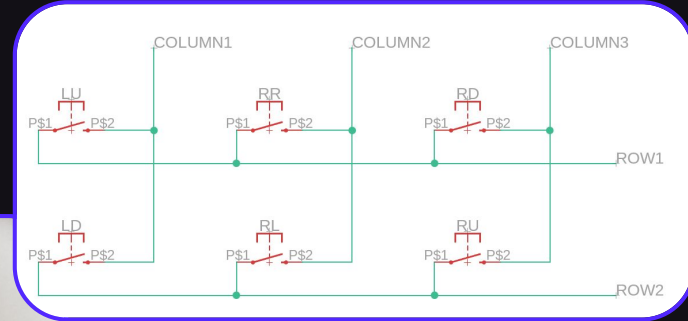
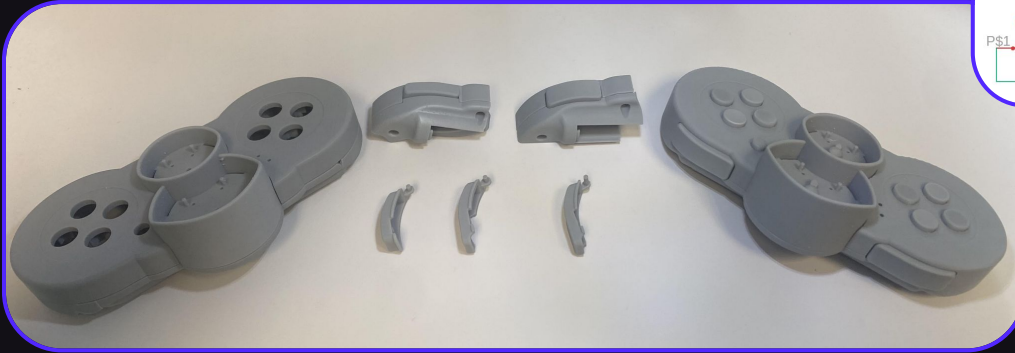


Step 2: Put a strip of double-sided adhesive on the plastic sleeve, about 5 mm from the edge.



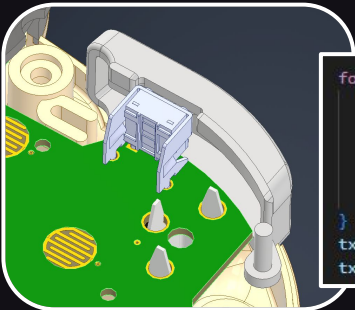
Flying Car Remote

Integrated a bumper system to the remote allowing for more possible player inputs for the finalized version of the game.



Learned how to fully implement a project into an existing system by:

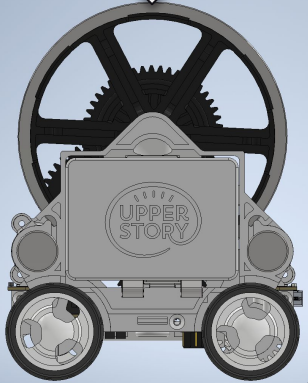
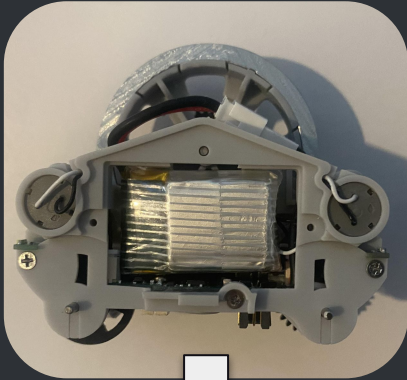
- Prototyping the mechanical solution and iterating
- Develop a working schematic to simulate the system
- Update and find new electrical components for the PCB
- Execute the changes in the code



```
for(int r = 0; r < NUM_ROWS; r++){
    GPIO_write(index: rowPins[r], value: 0);
    for(int c = 0; c < NUM_COLS; c++){
        buttons |= !GPIO_read(index: columnPins[c]) << buttonValues[r][c];
    }
    GPIO_write(index: rowPins[r], value: 1);
}
txData[HDR_LEN] = buttons;
txData[HDR_LEN + 1] = buttons2;
```

Removable Battery Solution

*this will change later

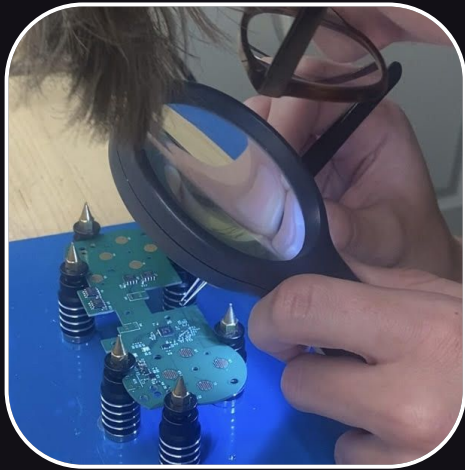


Designed a solution to enable non-stop play for the user by replacing batteries. The enclosed battery uses a custom PCB to connect to the inside of the car and battery charger.



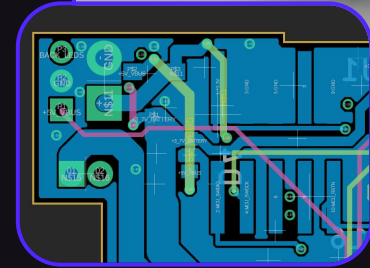
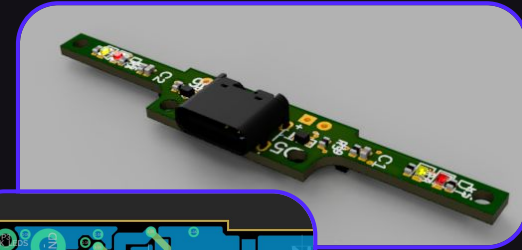
Learned how to work alongside someone on the same design allowing for changes to the battery and car frame to be cohesive and collaborative.

Printed Circuit Boards



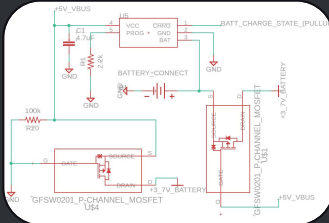
Learned how to use Fusion 360s PCB editor to make custom PCB changes such as:

- Remote bumper changes
- Car battery health improvements
- Battery connection board
- Charging station first iteration

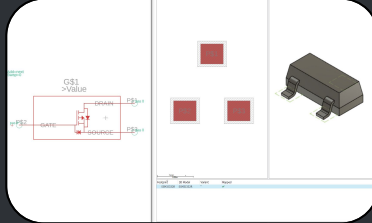


PCB Design Cycle

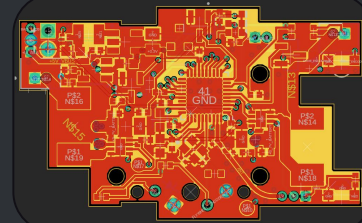
1: Build the schematic



2: Design part footprint



3: Route PCB layers



4: Produce circuit board

